

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Pearson Edexcel
International
Advanced Level

Centre Number

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Candidate Number

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Wednesday 23 October 2019

Morning (Time: 1 hour 30 minutes)

Paper Reference **WME01/01**

Mathematics

International Advanced Subsidiary/Advanced Level
Mechanics M1

You must have:

Mathematical Formulae and Statistical Tables (Blue), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$, and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for each question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

Turn over ►

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- (a) the common speed of the particles immediately after the string becomes taut, (3)
- (b) the magnitude of the impulse exerted on Q at the instant when the string becomes taut. (3)

Question 1 continued

Handwriting practice area with 30 horizontal lines.

(Total 6 marks)

Q1

Mark box for Q1



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

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Question 2 continued

Handwriting practice area with 30 horizontal lines.

(Total 9 marks)

Q2

Marking box for Q2



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DO NOT WRITE IN THIS AREA

- Given that the tension in the towbar is 2060 N, find the value of D .

DO NOT WRITE IN THIS AREA

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blank

Question 3 continued

Handwriting practice area with horizontal lines.

(Total 7 marks)

Q3



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 4 continued

DO NOT WRITE IN THIS AREA

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Leave
blank

Question 4 continued

Lined area for writing the answer to Question 4.

(Total 12 marks)

Q4



A diagram of a rectangular field. The bottom-left vertex is labeled A , the bottom-right vertex is labeled B , and the top-right vertex is labeled C . A horizontal dimension line below the field indicates the length AB is 5 m . A vertical dimension line to the right of the field indicates the width BC is 1.5 m .

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 5 continued

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Question 5 continued

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(Total 8 marks)

Q5



- (2)

Question 6 continued

Handwriting practice area with 25 horizontal lines.

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Question 6 continued

DO NOT WRITE IN THIS AREA

Question 6 continued

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DO NOT WRITE IN THIS AREA

Q6

(Total 12 marks)



7. Two forces, **F** and **G**, act on a particle. The force **F** has magnitude 4 N and acts in a direction with a bearing of 120° and the force **G** has magnitude 6 N and acts due north.

Given that $\mathbf{P} = 2\mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of **P**
(ii) the direction of **P**, giving your answer as a bearing to the nearest degree.

(7)

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Question 7 continued

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DO NOT WRITE IN THIS AREA

(Total 7 marks)

Q7



- (c) Find the time, to the nearest minute, at which the distance between the boats is the same as it was at noon.
- (8)**

Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

Q8

(Total 14 marks)

TOTAL FOR PAPER: 75 MARKS

END

